Transformations and Inverse Relations

Concept

Transformations move or scale a graph. Inverses swap its inputs and outputs.

Point mapping

For $y = a\,f(b(x-h)) + k$: $(u,v) \to (u/b+h,\ av+k)$, $a,b \neq 0$.

Inside change

Factor first: $f(2x - 6) = f(2(x - 3))$; shift right 3.

Inverse

Swap x,y and solve; use the horizontal-line test or restrict domain.

Domain / range

Transform allowed coordinates; inverse swaps domain and range.

Example 1

Map $(2,3)$ under $y = -2f(3(x - 4)) + 1$.

Solution

1. New $x = 2/3 + 4 = 14/3$.
2. New $y = -2(3) + 1 = -5$.
3. Mapped point: $(14/3,-5)$.

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Example 2

Find the inverse of $f(x) = (x - 2)^2 + 1$, restricted to $x \geq 2$.

Solution

1. Swap variables: $x = (y - 2)^2 + 1$.
2. Since $y \geq 2$, take the positive root: $y = 2 + \sqrt{(x - 1)}$.
3. $f^{-1}(x) = 2 + \sqrt{(x - 1)}$; domain $x \geq 1$, range $y \geq 2$.

Example 3

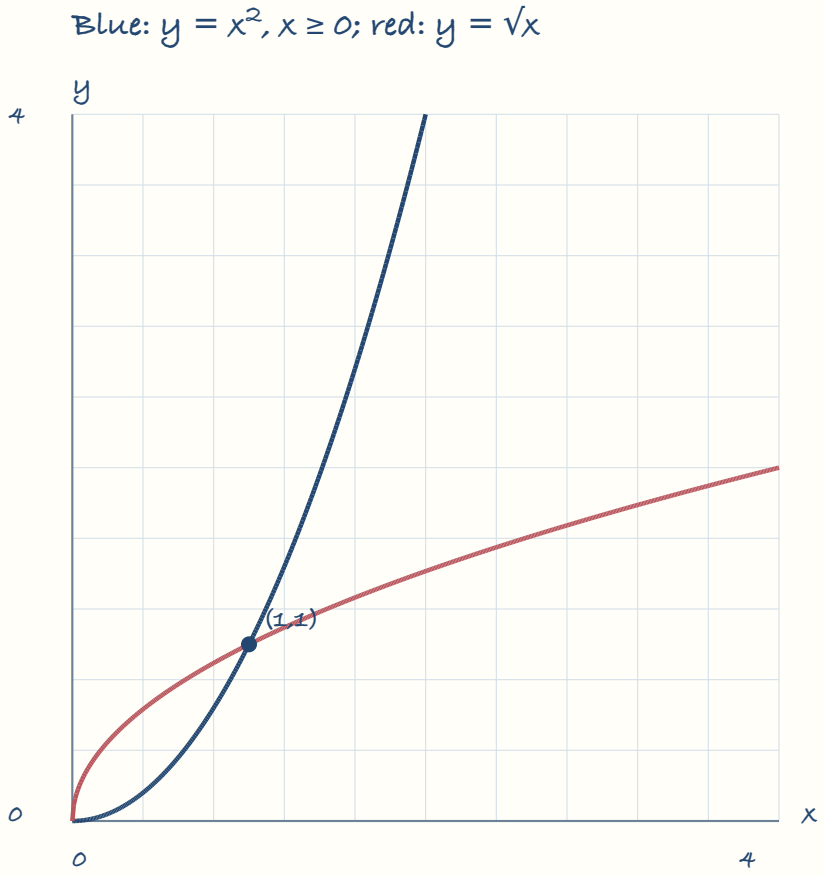
Describe $y = |x - 2| - 3$.

Solution

1. Move $|x|$ right 2 and down 3: vertex $(2,-3)$.
2. Domain all real x; range $y \geq -3$.
3. Piecewise: $-x - 1$ for $x < 2$; $x - 5$ for $x \geq 2$.

Transformations and Inverse Relations

Visual notes | read the graph



Equal axis scales show reflection across $y = x$. The restricted parabola and square-root graph are inverse functions.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Transformations and Inverse Relations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Map $(4,-1)$ under $y = 3f(2(x + 1)) - 2$.

2. State domain and range of $y = 2\sqrt{(x + 3)} - 4$.

3. Find the inverse of $f(x) = (x - 5)/2$.

4. Why is $y = x^2$ not one-to-one on the whole real line?

Transformations and Inverse Relations

Practice answers | check after trying

1. $(4/2 - 1, 3(-1) - 2) = (1,-5)$.

2. Domain $[-3,\infty)$, range $[-4,\infty)$.

3. $f^{-1}(x) = 2x + 5$.

4. Opposite inputs such as 2 and -2 give the same output 4, so the inverse relation would have two outputs for input 4.